1. All questions are compulsory.
2. Illustrate your answers with neat sketches wherever necessary.
3. Figures to the right indicate full marks.
4. Use of non-programmable calculator is permissible.
5. Assume suitable data if necessary.
6. Preferably, write the answers in sequential order.

Q1	**Attempt any Five (2x5=10)**	COs	Level
a)	Define Machine Learning.	CO1	1R2
b)	List any four techniques for handling missing values from datasets.	CO2	2R2
c)	State Principal Component Analysis in ML.	CO4	4R2
d)	State Hyperparameter Tuning of Model.	CO5	5R2
e)	List four applications of ML in the Healthcare domain.	CO6	6R2
f)	State the applications of the given libraries: Scikit-learn, Pandas, Numpy and Matplotlib	CO1	1R2
g)	State Mean Squared Error and Mean Average Error with importance in ML.	CO3	3R2
Q2	**Attempt any Three (4x3=12)**		
a)	Compare Supervised with Unsupervised Learning.	CO1	1U4
b)	Describe Precision and Recall with formula.	CO3	3R4
c)	Write a Python program to convert categorical columns (e.g., "Male/Female") into numerical format using one-hot encoding and label encoding.(Assume any tabular example)	CO2	2A4
d)	Write a Program to cluster given data into two groups using K-Means clustering assuming initial centroid as M1=4 and M2=11 Data points= {2,4,10,12,3,20,30,11,25}	CO4	4A4
e)	State PCA and K-Means clustering with example.	CO4	4R4
Q3	**Attempt any Three (4x3=12)**		
a)	Compare Linear Regression with Logistic Regression.	CO3	3U4
b)	Compare Grid Search and Random Search for hyperparameter tuning of model prediction.	CO5	5U4
c)	Write a Python Program to calculate MSE and R^2 for given values. Actual: 10, 12, 15, 18 Predicted: 8, 11, 14, 20	CO3	3A4
d)	Compare Bias and Variance of Machine Learning.	CO6	6U4
e)	Compare the Decision Tree with the Random Forest.	CO3	3U4
Q4	**Attempt any Three (4x3=12)**		
a)	Compare overfitting and underfitting in machine learning models.	CO2	2U4
b)	Describe the K-fold cross-validation technique for model evaluation.	CO6	6U4
c)	Write a program to perform Customer segmentation using the K-Means Algorithm. (Assume suitable data).	CO4	4A4
d)	Write a Python code to represent the given data using one-hot	CO2	2A4

	encoding. data = {'Fruit': ['Apple', 'Banana', 'Cherry', 'Apple', 'Cherry', 'Banana']}		
e)	Describe standardisation and Normalisation techniques of feature scaling with an example.	CO2	2U4
Q5	**Attempt any Two (6x2=12)**		
a)	Write a program to implement PCA on the Iris dataset (4 features) to reduce it to 2 components and visualise the results.	CO4	4A6
b)	Solve the given problem mathematically Study Hours: 2, 3, 5, 7, 9 Exam Marks: 50, 60, 75, 85, 95 1. Calculate Slope and Intercept. 2. Predict marks for students who study 6 hours. 3. Write a Python program for the above.	CO3	3A6
c)	Write a program to tune hyperparameters using Grid search and Random search which going to train on Decision tree classifier. data = {'Age': [25, 30, 45, 22, 35, 50, 40, 29, 55, 42], 'Income': [50000, 60000, 80000, 45000, 75000, 90000, 82000, 54000, 100000, 85000], 'Loan_Amount': [2000, 4000, 10000, 1500, 5000, 12000, 9000, 3000, 15000, 11000], 'Credit_Score': [650, 700, 600, 720, 710, 580, 690, 660, 560, 640], 'Default': ['No', 'No', 'Yes', 'No', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes'] }	CO5	5A6
Q6	**Attempt any Two (6x2=12)**		
a)	Write a program to Split given tabular data into train, test and validation. ID: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 Age: 22, 25, 47, 52, 46, 56, 55, 60, 23, 40 Income: 35000, 40000, 52000, 60000, 48000, 70000, 71000, 65000, 35000, 45000 Purchased: No, No, Yes, Yes, Yes, Yes, Yes, Yes, No, Yes	CO5	5A6
b)	Write a program to represent given data using Hashing encoding technique. data = [{'fruit': 'apple'},{'fruit': 'banana'},{'fruit': 'cherry'},{'fruit': 'apple'},{'fruit': 'banana'}]	CO2	2A6
c)	Solve the given Logistic regression problem mathematically. Consider threshold 0.5 and e=2.71828 Email: 1, 2, 3, 4, 5, 6 "FREE" Count: 3, 0, 1, 4, 2, 5 "Links" Count: 5, 1, 0, 7, 2, 8 Spam (y): 1, 0, 0, 1, 0, 1	CO3	3A6

Course Outcomes:

CO1: Use of Machine Learning (ML) tools and libraries.

CO2: Transform and normalise the datasets by encoding features.

CO3: Implement Supervised Learning Algorithms using Python.

CO4: Implement Unsupervised Learning Algorithms using Python.

CO5: Evaluate the ML model performance using appropriate metrics.

CO6:Solve the real-world Machine Learning problems.